AI Alignment and Governance as Normative Institutional Design

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Technical approaches to AI alignment and governance have increasingly drawn on traditions developed for problems in moral philosophy, political theory and legal systems—from constitutionalism and social choice, to deliberative democracy and virtue ethics. However, technical alignment methods draw on these traditions unevenly. Some traditions, such as constitutionalism, have inspired substantial bodies of technical work, while others remain largely unexplored. Moreover, alignment methods often carry over only a fraction of the concepts that make up the traditions they draw on. In order to provide a systematic understanding of this landscape, we construct a property space that places AI alignment and governance approaches into fifteen regimes, records who holds authority over their norms, and maps each approach to the traditions in philosophy, political theory, and legal theory that it draws on. We use this property space to study *normative translation*—how methods carry over the concepts of the traditions they draw on—analyzing which concepts are preserved, partially preserved, or omitted, and what consequences these choices bring about. We highlight a recurring tendency of alignment methods to treat normative disagreement as a problem to be *resolved away*, even when optimizing for incommensurable objectives and drawing from underlying traditions that argue against such reduction. Finally, mapping the full landscape reveals underexplored regimes and omitted concepts. We argue that these gaps point toward different goals for AI alignment: for example, not invisibly collapsing disagreement of norms, but building methods that can hold and contest it.

Keywords: AI Alignment, AI Governance, Moral Philosophy, Political Theory, Legal Theory, Value Pluralism, Institutional Design

AI Alignment & Governance

1. Introduction

What should an AI system do when the values it is meant to follow conflict? How should human preferences that differ across individuals determine its behavior? Should those preferences be aggregated into an objective, encoded as explicit principles, or resolved through deliberation? And when reasonable people continue to disagree, should an aligned system resolve it at all, or instead preserve and govern disagreement over time? These questions are increasingly central to AI alignment and governance, but they are not uniquely questions about AI. Subfields of philosophy have long studied how competing values and obligations should be weighed (Berlin, 1969; Ross, 1930); political theory has asked how institutions can govern populations that disagree about what is good or just in a legitimate and stable fashion (Habermas, 1996; Rawls, 1993); and legal jurisprudence has developed systems for specifying and contesting norms over time, as societies evolve (Dworkin, 1986; Hart, 1961). These traditions have been realized in concrete institutions: from codified rules in constitutions to democratic procedures aggregating competing preferences; resulting in different institutional arrangements for operating and governing under disagreement.

In recent years, the field of AI alignment has begun to draw on these traditions. For example, Constitutional AI (Bai et al., 2022) encodes normative principles in an explicit natural language artifact that humans can understand for AI systems to follow (often called "a constitution", "model spec", or "code of conduct"); social-choice approaches to preference learning aggregate judgments according to individuals' preferences or incentives (Ge et al., 2024), and utility-theoretic ones combine several reward dimensions (Maura-Rivero et al., 2025); deliberative approaches derive principles through collective

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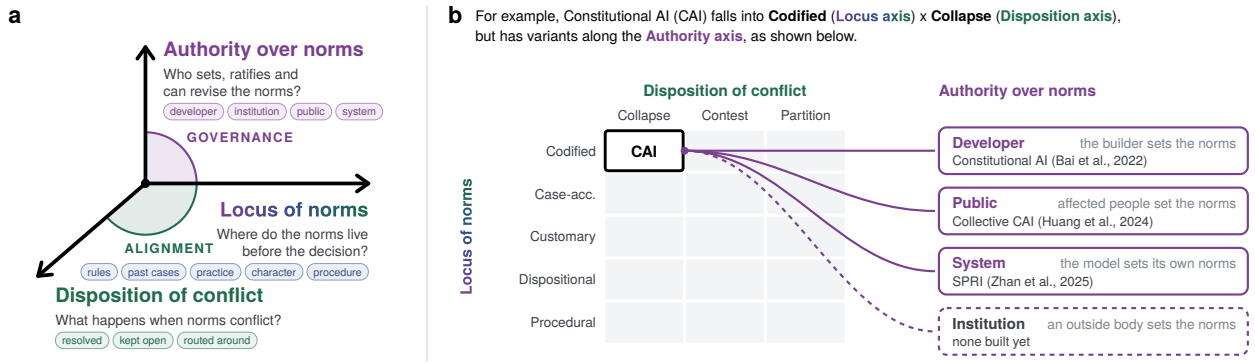


Figure 1 | **Overview of the Normative Design Space.** (a) Three axes: who has *authority over norms* (governance), where the *locus of norms* is specified (shared by governance and alignment), and what the *disposition of conflict* is (alignment). (b) A method largely fixes its locus and disposition but leaves authority open. Constitutional AI falls into Codified and Collapse, while its variants ~~place authority with~~take their norms from the developer (Bai et al., 2022), the affected public (Huang et al., 2024), or the system itself (Zhan et al., 2025). No Constitutionalism method in the encyclopedia ~~places it with~~takes them from an external institution.

procedures (Huang et al., 2024); and pluralistic systems attempt to represent information and ultimately respond in line with heterogeneous and sometimes directly opposing values (Feng et al., 2024; Sorensen et al., 2025). As a result of the success of these technical approaches, concepts originating in moral philosophy, jurisprudence, and political theory have become a regular part of the technical vocabulary of AI alignment and governance.

Yet, this transfer of concepts is often selective. For example, Constitutional AI (Bai et al., 2022) draws on constitutionalism but carries over only some of its concepts, as we illustrate in Section 4. Some simplifications follow from engineering or infrastructural constraints; others are design choices that shape the resulting system. Our broader concern with such transfer is that the **normative behavior of a system can change when a concept is separated from the institutional context in which it performed its function**. We call the process by which a method carries over the concepts of a tradition *normative translation*, and in this paper, ask what consequences follow when concepts are only partially preserved in translation.

In order to study this, we systematically map AI alignment and governance approaches to the traditions in political theory, legal theory, and philosophy that they draw on, by asking which regime each approach falls in. Following the typology construction of Collier et al. (2012), we organize approaches according to three recurring dimensions: where norms reside, what happens when norms conflict, and who holds authority over them (Figure 1). Mapping existing approaches into this space reveals an uneven landscape: while alignment systems increasingly incorporate heterogeneous sources of normative input, comparatively few preserve disagreement as an enduring feature of the resulting system. This allows us to understand not only which normative concepts have influenced technical alignment systems, but the implications of how these have been translated and which regions of the design space remain underexplored.

Our contribution here is threefold. First, we survey AI alignment and governance approaches that draw on normative traditions. We provide a searchable encyclopedia of these traditions, their concepts, and the approaches that draw on them to aid researchers in work involving these concepts. Second, we construct a property space along three dimensions: the locus of norms, the disposition of normative conflict, and authority over norms. Methods largely fix the first two, which place them into regimes, and leave the third open. This, in turn, allows us to discuss

the *coverage* of the space. Third, we analyze how existing methods translate the concepts of the traditions they draw on, distinguishing shortfalls against concepts a method claims from omissions of concepts it never set out to carry over, and the potential consequences of each. Our goal is not to critique AI systems for not reproducing a tradition identically, but rather, to argue that drawing on a tradition should make its institutional assumptions explicit. By understanding the space of concepts holistically, and identifying the gaps in current alignment methods, we hope to identify both the strengths and limitations of existing approaches, as well as provide a systematic basis for exploring alternatives for future systems.

2. Related Work

Normative theory in AI alignment. A growing body of work draws explicitly on normative concepts from philosophy, political theory, and jurisprudence (Dobbe et al., 2021; Gabriel, 2020; Zhi-Xuan et al., 2025). Examples include pluralistic alignment (Sorensen et al., 2025), social choice for aggregating heterogeneous human preferences (Conitzer et al., 2024; Maura-Rivero et al., 2025), deliberative democracy for collective alignment (Ovadya et al., 2025; Tessler et al., 2024), constitutionalism for principle-based alignment (Bai et al., 2022; Huang et al., 2024), and jurisprudential concepts such as precedent and legal interpretation for case-based alignment and specification design (Caputo, 2025; Kolt et al., 2026). These works establish individual connections between normative traditions and technical mechanisms; we provide a broader mapping of these connections in Appendix A.

Typologies and property spaces in political and social sciences. Our approach follows the tradition of typological analysis in political and social sciences, where multidimensional property spaces are used to form concepts and distinguish theoretically meaningful types (Barton, 1955; Collier et al., 2012; Elman, 2005; George and Bennett, 2005). This process decomposes variation into underlying dimensions whose cross product constitutes a property space, and thereby also exposes combinations that are not reflected by existing classifications. Importantly, cells in this property space which are theoretically possible but unoccupied can themselves be analytically informative.

Existing surveys of AI alignment and governance. Recent surveys of AI alignment are primarily focused on individual subcategories of technical approaches (e.g., value pluralism) and accordingly organize systems according to technical mechanisms. For example, Sorensen et al. (2024) distinguish Overton, steerable, and distributional pluralism, while other frameworks distinguish approaches according to how preferences or perspectives are represented and aggregated. We instead construct a *property space* (Collier et al., 2012) from recurring dimensions of normative institutions, described next.

3. A Normative Design Space for AI Alignment and Governance

In order to map the space of AI alignment and governance and study how traditions are drawn on across fields, we create a property space defined by three dimensions: the locus of norms, the disposition of normative conflict, and authority over norms. We derive these dimensions through a systematic mapping of AI alignment and governance literature followed by conceptual substruction (Barton, 1955): the "procedure of finding, for a given system of types, the attribute space in which it belongs and the reduction which has been implicitly used" (Lazarsfeld, 1937, p. 132). Overlapping surface categories are, therefore, decomposed into the underlying attributes that distinguish them.

Whereas the functional reduction of prior work (Lazarsfeld, 1937, pp. 127–128) removes combinations that do not occur in practice, we retain them, since we are interested in gaps in the literature: an empty or sparsely populated cell points to an underexplored architecture domain. Our approach also differs from existing taxonomies of AI alignment, which group approaches according to their surface-level technical mechanisms within tradition-derived categories, in that

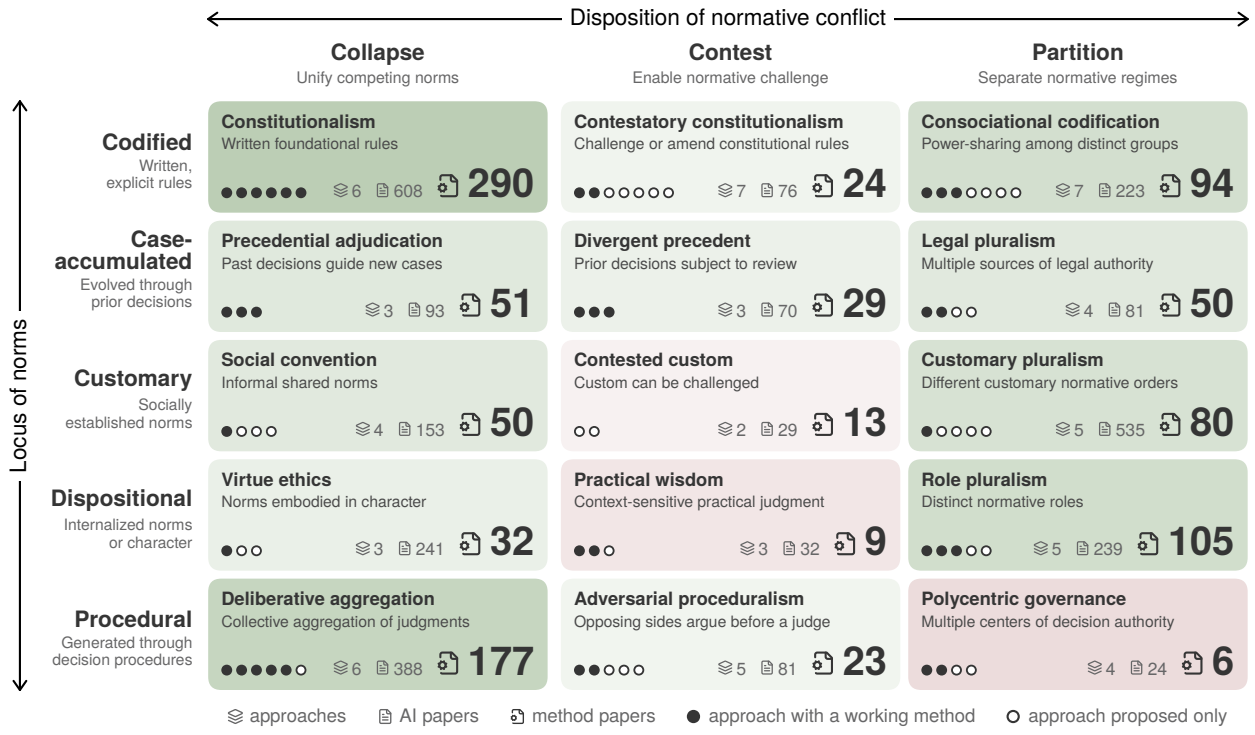


Figure 2 | **The Normative Design Space for AI Alignment and Governance.** The two axes, *locus of norms* (rows) and *disposition of normative conflict* (columns), yield fifteen *regimes* (cells). The third dimension, authority over norms (Figure 1), varies within regimes and is reported in Section 3.3. A short gloss explains each row, column and cell. Each cell contains the respective regime's name, a short description, and counts of approaches and papers (within AI literature) that fall under it; approaches are counted in the encyclopedia, papers and method papers in the corpus (Appendix B). Each dot corresponds to one approach—filled where at least one working method in the encyclopedia exists, empty where the approach has only been proposed. The large number in the bottom right corner of each cell counts method papers, which is a subset of papers that explicitly specify a technical mechanism. The background color corresponds to the log of method paper count. The counts cannot include methods built in industry but never released as papers, or papers that our search missed (Section 6). They reflect published and publicly released work, and should be read as the relative sizes of the regimes rather than as exact counts.

it places methods rather than traditions. A tradition takes positions on all three dimensions, and its proponents often disagree on each. We instead place each method by the defining features of its implementation, which allows methods from different traditions to be compared. The focus on normative conflict is our choice, and it shaped both our search and our keywording. Other features (e.g., transparency or accountability) would yield a different space, with different gaps. We choose this one because the traditions that alignment draws on were developed in large part to govern disagreement (Section 1), and because coders place methods in it with substantial agreement (Appendix B.9).

To construct the space, we first collect an initial sample of works (Appendix B.1) through keyword searches and the reference lists of three surveys (Ji et al., 2023; Lu et al., 2025; Shen et al., 2023), and draw up surface categories from the normative traditions they draw on. Per Petersen et al. (2008), we then extract the normative concepts and keywords of each record. Unified across studies, the keywords show where the surface categories overlap, and substruction decomposes them into the attributes that distinguish them. We populate the space through a systematic search of

arXiv and OpenAlex, extended by three rounds of citation snowballing. Our corpus contains every work found that proposes, builds, evaluates or analyzes a mechanism determining which norms an AI system follows when more than one norm, value or stakeholder bears on a decision, or how it handles conflict among them (3,056 records, 1,033 of them methods). A purposive subset of 192 records, which we annotate in depth and group into 67 approaches, forms the encyclopedia we release with this paper. Full collection and inclusion criteria, as well as statistics, are provided in Appendix B.

This yields three dimensions: the **locus of norms**, which indicates where the norms that bind a decision reside prior to the moment of decision (not who holds authority over them), the **disposition of normative conflict**, which indicates what happens when those norms conflict—whether the conflict is resolved, kept open, or routed around, and the **authority over norms**, which indicates who may write, interpret and change them. The full property space is 5 x 3 x 4; we call the fifteen cells of its first two dimensions *regimes*, and since most methods leave authority open, we report it within each regime (Section 3.3). We define the terms used throughout the paper in Appendix C.

We also considered the stage at which norms act (training, inference, or post hoc), but as a property of a method's pipeline rather than of a tradition, it distinguishes none of the surface categories; instead, we code dispositions across all stages of a method (Section 3.2). Appendix B.5 argues for each category within the dimensions: the traditions it comes from, the question that separates it from its nearest neighbor, and what merging the two would lose.

3.1. Locus of Norms

The *locus* of a norm is where it resides before the decision it binds is made. For the locus of norms, we find the following categories:

1. **Codified.** Norms are stated as explicit rules in a written specification that is set before any particular case arises (Hart, 1961).
2. **Case-accumulated.** Norms reside in a growing body of past decided cases. No case is decided by a rule stated in advance, but by what accumulated decisions have held (Jonsen and Toulmin, 1988; Levi, 2013).
3. **Customary.** Norms reside in regularities of practice. Unlike codified or case-accumulated norms, they are never authoritatively stated; they are learned through participation and enforced by expectation rather than by rule or ruling (Burke, 1987; Lewis, 1969).
4. **Dispositional.** Norms reside in the system's character rather than in any external artifact. There is no rule to consult and nothing to look up, only evidence of character traits of the system itself (Aristotle, 1925).
5. **Procedural.** Norms do not persist between decisions. What should be done is whatever a specified procedure (e.g., a vote, a deliberation, a debate) yields at the moment of decision (cf. Habermas, 1996).

The categories are separated by three questions. First, does the norm exist before the decision is made? If not, the locus is procedural. Otherwise, is there an artifact to consult? If so, the locus is codified when that artifact is a written specification set in advance and case-accumulated when it is a record of decided cases. If not, the locus is customary when the norm is sustained by many agents' expectations of one another (Lewis, 1969), so that any agent can contest it by acting otherwise, and dispositional when it is part of a single system's character (Aristotle, 1925), so that disagreement can surface only within that system (Appendix B.5).

3.2. Disposition of Normative Conflict

A *normative conflict* arises when two or more norms that apply to a case recommend incompatible actions, whether these are values held by one agent or views held by different people. It differs from *normative ambiguity*, where the applicable norms leave the right action open. Ambiguity is filled in by the locus (e.g., by precedent, custom, the system's character, or a procedure), and so does not by itself call for a disposition. For

the disposition of normative conflict, we find the following categories[1]:

1. **Collapse.** The conflict is settled by a ranking (e.g., a scalar objective, a weighted sum, or a priority order among principles) and the losing norms have no further force (i.e., nothing lets them reopen the decision or bear on later ones).[2]
2. **Contest.** The conflict is preserved and the losing norms retain force. This means that the output either holds the competing norms open for a later decision (e.g., by setting out the positions, not choosing among them, and deferring to the user) or presents a standing channel (e.g., a mechanism for appeal or amendment), based on which the case itself can be reopened or have an impact on later ones (Mouffe, 2000; Pettit, 1997).
3. **Partition.** The conflict is avoided. Several normative orders are kept intact, each authoritative within its own domain, and a rule sorts cases among them by features of the case (e.g., who is asking, where, or in which domain) (Lijphart, 1977; von Savigny, 1880; Walzer, 1983).[3]

The categories are separated by two questions. First, is the conflict sorted to one of several normative orders by features of the case, without ranking the orders? If so, the disposition is partition. Otherwise, once the system acts, can the losing norm still change any one of the following: a decision the output leaves open, the decision itself on appeal, or later decisions of the system? If so, the disposition is contest; if not, collapse. We code dispositions at the level of the whole method, across both training and inference, since force is often removed in inference. For instance, a reward model utilizes minority judgments at training time but absorbs them into a single score at inference time. Conversely, aggregation at some step of the pipeline, which nearly every learning system performs, does not by itself make a method collapse: what decides is whether the losing norm keeps force once the system acts (e.g., through an appeal channel). Where the stages of a method fall in different regimes (e.g., data collection, training and deployment, as when a public deliberates on principles that are then codified and trained into a model), we place the method by its own contribution and record the other stages as secondary placements (Appendix B.6).

Disposition and incommensurability. The three disposition categories differ in what they presuppose about the ability to rank conflicting norms against one another. Two values are *incommensurable* when no common measure ranks their realizations; neither option is better, yet they are not equally good, as a small improvement to one would not necessarily make it better than the other (Raz, 1986; cf. Chang, 2002, who reads the same pattern as parity rather than incomparability; see Andersson and Hsieh, 2025 for a more thorough overview).[4] Collapse ~~presupposes~~usually works through a complete ordering of the options—whether achieved through a weighted sum of component values (see Anderson, 1993 and Sunstein, 1994 for critiques) or a strict priority among principles—under which the

[1] We considered open, axial and selective coding (Strauss and Corbin, 1998), as well as the first-order, second-order and aggregate-dimension ladder (Gioia et al., 2013), but note that the keyword-based approach is more native to the computer science literature it addresses. Alternatives would likely yield a comparable outcome in terms of the intermediate categories.

[2] Practically, most existing systems that fit this category resolve the conflicting norms into a single output, without trace or acknowledgment of the disagreement that produced it. A system that acknowledges them can still collapse if the acknowledgment is not paired with potential further force ascribed to the losing norm. Such a system resembles the "much more disaggregated accounting" that Sunstein (1994, p. 843) recommends where regulation would otherwise be judged on a single metric, one that "exposes to public view the full set of effects"; in our terms, this still collapses.

[3] Partition does not eliminate conflict but relocates it to cases in between norms or precedent. This is where the sorting rule decides which norm takes priority, which is a problem known in law as choice of law (von Savigny, 1880) (cf. de Sousa Santos, 2002 on interlegality).

[4] We use *incommensurable* for abstract values (e.g., helpfulness and harmlessness) and *incomparable* for the options that bear them (e.g., two candidate responses). Usage in the literature varies (Andersson and Hsieh, 2025; Chang, 1997). Incommensurability is one ground for resisting collapse; reasonable disagreement among persons, which applies to both commensurable and incommensurable values, is another (Rawls, 1993).

rejected options are plainly worse, although a system can also choose among incomparable options and still collapse by foreclosing review. Where values are incommensurable, no such ordering exists. And so, if a decision is still made, it leaves a remainder (Williams, 1965, 1981), a loss that the chosen option does not cancel and that collapse erases. This distinguishes *optimization*, which picks an option that is at least as good as every other, and hence needs a complete ordering, from *maximization*, which picks an option that no other option beats, and hence tolerates incomparable ones (Sen, 1997). A system that maximizes may keep a record of the unbeaten options open for later, which would be one form of contest. Contest and partition draw on two prominent responses to such value conflict in the pluralist tradition. Contest treats this conflict as a permanent condition to be governed rather than settled once and for all (Berlin, 1969; Mouffe, 2000; Pildes and Anderson, 1990), and aims to keep it constructive rather than destructive (Deutsch, 1969; Stray et al., 2023). Partition rejects the comparison altogether, applying different decisions in different spheres, governed by their own norms (Anderson, 1993; Walzer, 1983).

3.3. Authority over Norms

The *authority* over a norm is who may write, interpret and change it. These powers can sit with different actors: those who supply a norm (e.g., participants who propose principles) need not be those who adopt, interpret and revise it (often the developer). It is distinct from the locus: two systems can follow the same constitution and differ only in this respect, for example when one is maintained by its developer and the other is overseen by an independent body. For authority over norms, we find the following categories:

1. **Developer.** The organization that builds or deploys the system sets and changes its norms, much as the designer of a mechanism sets the rules under which others act (Hurwicz, 1973) and those who write code govern its users (Lessig, 1999) and platforms govern their users' speech (Klonick, 2018).
2. **External institution.** A body outside the developer (e.g., a legislature, a court, or a regulator) sets the norms for the system to follow. Its authority is derived from a legal or institutional order (Hart, 1961), as in proposals to align systems to law (Kolt et al., 2026).
3. **Affected public.** The people affected by the system (e.g., its users, a community, or a sample of the public) set its norms, on the principle that those affected by a decision should have a say in making it (Goodin, 2007; Habermas, 1996).
4. **System itself.** The system, or a population of AI agents, generates or revises its own norms, as norms arise from the practice of those they bind in custom and convention (Hayek, 1973; Lewis, 1969), or from an agent's own practical judgment (Aristotle, 1925).

The categories are separated by two questions. First, are the norms set by people or by the agents they bind? If by the agents, authority lies with the system itself. Otherwise, where does the authority of those who set them come from: from building or deploying the system (developer), from an office outside the developer (external institution), or from being affected by the system (affected public)? The answers matter because they fix who can hold the system to its norms, and on what claim (Appendix B.5).

Unlike locus and disposition, authority is rarely fixed by a method. Constitutional AI, for instance, works the same whether ~~its constitution is written~~the principles of its constitution are supplied by the developer (Bai et al., 2022), by a sample of the public (Huang et al., 2024), or by the model itself at inference time (Zhan et al., 2025) (Figure 1b). We therefore code authority for each method paper as instantiated in its experiments, i.e., by who supplies the norms, and record separately whether the method fixes it, i.e., whether its contribution depends on who supplies the norms (e.g., a procedure for public input) or would run unchanged with norms from another source. Because most methods do not specify authority over norms, we place methods in regimes by locus and disposition, and, where applicable, report authority options within each regime.

Among the surface categories, only mechanism design places authority with the designer; the others place it with institutions, with the public, or with those the norms bind (Table 8). Among the methods in the encyclopedia, by contrast, 65 of 80 leave authority open, and in their experiments it is most often held by the developer (41) and least often by an external institution (5; Table 21), whose norms (e.g., statutes or court decisions) the methods draw on but do not let it revise. The methods that do fix authority place it with the affected public or with the system itself. ~~Placing it with the public~~Having the public supply the norms does not by itself keep disagreement open: 65% (17 / 26) of the methods whose experiments ~~give authority to~~source their norms from the affected public still collapse conflict.

Figure 4 in Appendix B.10 traces how the number of methods for each locus and each disposition has grown over time. Figure 3 breaks each regime's literature down by the kind of contribution it makes.

4. Case Studies of Normative Translation in Alignment

To understand how normative concepts are translated into technical systems, we study three representative alignment methods as case studies. We chose these methods because they are widely adopted as alignment techniques. For each approach, we examine (a) its regime within our taxonomy, (b) the traditions it draws on and the concepts that make up these traditions, (c) whether the approach preserves, partially preserves, or omits these concepts, and (d) how the conceptual differences may affect the resulting regime and people's expectations of it. Throughout, we distinguish the connections that the source works claim themselves from those that we identify. Across the encyclopedia, translation is rarely complete. Of the 96 concepts that the approaches' own sources claim or imply, only 10% (10 / 96) are fully carried over, and a further 102 concepts are not claimed at all (Table 17).

4.1. Constitutional AI and Constitutionalism

Constitutional AI (Bai et al., 2022) and the methods that build on it draw on the normative concept of **constitutionalism**. In these methods, an explicit set of principles that a model should follow is codified in a single artifact, often called the constitution or the model specification (Anthropic, 2026a; OpenAI, 2026). This artifact, framed as an authoritative source of guidance, is then used to guide and/or evaluate the model's behavior. In our taxonomy, this instantiates the regime of **codified norms** and **collapsing conflict**.

Authority over these norms lies with whoever writes the constitution. In the original method, this is the developer (Bai et al., 2022); in Collective Constitutional AI, ~~it is~~the norms are supplied by a sample of about 1,000 U.S. adults, who proposed and voted on statements that the developer then turned into a constitution (Huang et al., 2024). The developer, however, still decides what enters the text, what counts as compliance with it in training, and whether it is later revised. In both cases, the alignment method is the same, while ~~the authority over norms~~who supplies the norms differs, highlighting the flexibility of Constitutional AI to various governance methods.

In legal and political theory, constitutionalism involves more than the codification of rules, however. It couples the rules with frameworks to interpret, apply, contest, and revise them (Dworkin, 1986; Hart, 1961). Constitutional AI preserves one part—explicit codification—but only partially the other one—interpretation of the rules~~, as models apply the codified rules in different contexts without clear guidelines~~. Developers do interpret their rules through annotation guidelines, behavioral policies and evaluation rubrics, but these are mostly internal, and those bound by the rules cannot contest them. In particular, ~~it omits what~~every rule has what Hart (1961) calls an open texture ~~of rules, as~~, and Constitutional AI leaves its borderline cases ~~have neither a procedure nor a designated decider~~with neither a designated decider nor a way to review how they were decided (Table 18).

Furthermore, institutional frameworks for revising the constitution do not exist. While devel-

opers do revise their model's constitution, it is not conducted through a defined procedure with precise mechanisms for precedent, appeal by stakeholders or the public, or oversight from external authorities. To our knowledge, there is also no channel through which an individual can contest a specific interaction against the constitution (for example, when they believe the interaction was not faithful to it) and have it reviewed. Revision and contestation are concepts of the neighboring regime of Contestatory constitutionalism, and we return to them in Section 5.3.

As a result, the normative structure of Constitutional AI is not consistent with constitutional governance as it is defined and conventionally understood. A fixed specification in isolation, without the surrounding institutional apparatus, separates the normative text from its realization. The translation, therefore, preserves the constitution as a source of principles, but omits most of the mechanisms through which constitutional authority is actually exercised, challenged, or amended. Since Bai et al. (2022) do not claim these concepts, they are omissions rather than shortfalls (Table 18).

We are not suggesting that this translation is necessarily defective or negative. Specifying norms explicitly in a constitution (using natural language as opposed to, say, a neural network-based adapter) makes normative guidance understandable to humans, even those who are not involved in the AI's development. However, it also significantly changes how normative authority is structured. In constitutional governance, authority is distributed across the text, its interpretation, precedent, and institutional roles. In AI constitutionalism, it becomes concentrated in the specification itself and in the process by which the specification is written.[5] The term *constitution* also makes the concept more familiar to laypeople, and it arguably carries a favorable connotation. This can help engage laypeople in AI governance. However, it can also create the misleading impression that AI constitutionalism affords the same transparency and mechanisms as political constitutions, which it does not.

4.2. Preference-based Alignment and Social Choice

Preference-based alignment uses human judgments to train reward models, which then steer the outputs of AI systems toward an approximation of aggregate preferences (Christiano et al., 2017; Ouyang et al., 2022). Conceptually, these ideas have roots in social choice theory, which asks how individual preferences should be aggregated into a collective decision (Arrow, 2012; Conitzer et al., 2024), and, more distantly, in utilitarianism (Bentham, 1996; Mill, 1998). However, most technical frameworks for preference optimization do not explicitly claim to implement these traditions. In our taxonomy, this instantiates the regime of **procedural norms** and **collapsing conflict**. Standard RLHF does not model the plurality of its annotators explicitly and, therefore, fails our plurality criterion (I2; Table 3). We thus code the approach that builds on it by writing an explicit voting rule into the training objective, with RLHF as its baseline (Table 18).

The authority over norms rests with the developer, who organizes the process of preference data collection. Specifically, the developer chooses the annotators, their instructions, and the aggregation rule (often implicit in the loss function). The judgments of these annotators are aggregated and conditioned on at training time, masking individual or minority preferences.

The technical translation preserves individual judgments at input time, as the comparative judgments of annotators are preserved in the data.[6] However, these judgments ultimately feed into a single reward function, ranking, or model policy, which represents only the aggregate of the population's choices. Furthermore, an annotator's preference among multiple options hides the reasons behind the choice and how strong the

[5]The process of writing a constitution is currently left largely to the discretion of its authors, who may, but are not required to, consult other stakeholders. To the best of our knowledge, no major model aligned through AI constitutionalism currently allows individuals outside the authoring organization to contest or directly amend its constitution.

[6]Concretely, the data are relative judgments, collected from annotators who are most often shown two or more model responses and asked which they prefer or to rank them (Christiano et al., 2017; Ouyang et al., 2022).

preference is (Hausman et al., 2017).

Preference-based alignment methods, therefore, represent plurality upstream (in data collection and training) but collapse it downstream (at inference time) and do not afford individuals authority over the deployed system's behavior. The translation, therefore, omits procedures that determine how preferences are weighed or how affected parties could challenge the learned objectives. From the standpoint of social choice, the aggregation rule itself is implicit and unchosen. Standard preference learning aggregates judgments much like a Borda count (Siththaranjan et al., 2024), a rule fixed by the choice of loss function rather than chosen and justified by the properties that a collective decision should satisfy (Ge et al., 2024).

One direct consequence of this translation is that plurality of judgments at the source may not result in plurality in the resulting system: to make preference optimization work at a large scale, these methods resolve diverse preferences into an aggregate policy. While important for reasons of computational feasibility, another consequence of this, primarily from a normative standpoint, is that the developer decides how preferences of a population are represented, preserved, and weighted on behalf of the population (annotators and users).

4.3. Collective Deliberation and Deliberative Democracy

Deliberation-based alignment methods explicitly incorporate people's input, as a form of democratic participation, to construct model specifications or to influence alignment training (Huang et al., 2024), as in Meta's Community Forums on generative AI (Meta, 2026). They draw from deliberative democracy, which gives individuals the opportunity to participate in decision making, as well as reasoning about and challenging collective goals (Habermas, 1996). In our taxonomy, this instantiates the regime of **procedural norms** and either **contest** or **collapsing conflict**. Where the procedure only supplies the principles of a written specification, as in Collective Constitutional AI, the method falls under codified norms instead (Section 3.2).

The authority over norms is meant to rest with the participating public. How much authority the public actually holds, however, depends on ~~how faithfully a method implements deliberation~~three separate questions: whose judgments are solicited, whether participants can exchange reasons and reconsider their views, and what they are empowered to decide. A group can deliberate seriously and still only advise the developer, and a vote can settle a decision without any deliberation. Some methods only collect the opinions of individuals (e.g., through votes on a set of statements). Others enable active participation, in which individuals discuss and give qualitative feedback that leads to a one-time revision of the model. Few, if any, let individuals revisit a decision once the system acts on it.

In most cases, the technical translation preserves the collective judgments of participants. However, it omits the institutions through which, in deliberative democracy, collective judgment remains authoritative over time. There are currently no frameworks that combine representative participation with a standing channel to contest decisions later, or that support deliberation at more than one stage of the alignment pipeline.

One direct consequence is that public participation does not, by itself, keep disagreement open. For example, Bakker et al. (2022) fine-tune a language model to write consensus statements for groups of participants with diverse opinions. A reward model predicts how much each participant would agree with each candidate statement, and the statements are then ranked by a social welfare function that the researchers choose. The group is left with one statement, and the views it does not reflect have no further effect. In the encyclopedia, 65% (17 / 26) of the methods ~~that place authority with~~whose norms are supplied by the affected public still collapse conflict (Section 3.3). Authority over norms and the disposition of conflict, therefore, need to be analyzed separately.

4.4. The Consequences of Translation Gaps

Across these examples, normative translation tends to preserve the concept of a tradition

that is most readily represented as a technical mechanism. Constitutions become specifications, heterogeneous preferences become aggregation problems, and deliberation becomes a procedure for producing judgments. The surrounding concepts, through which institutions manage interpretation, disagreement, revision, and competing authority, are preserved less consistently. As the case studies show, reducing a concept to its technical mechanism without the surrounding institutional structure shapes the normative architecture of the resulting system. In particular, the resulting system cannot represent authority or disagreement in the way the original concept does.

What follows when disagreement is not preserved? First, there is no recourse. A user or community whose experience is at odds with their expectations of the model's behavior has no channel through which to contest the decision and potentially reopen it. Minority positions may hence be overridden without a trace. Second, alignment errors (i.e., deviations from the correct behavior under the system's governing norms) persist until the developer notices them, as affected parties cannot trigger a correction. Third, the legitimacy of the system's behavior rests on the author of its norms, which is at present most often the developer.

Who could arbitrate this instead? The traditions suggest several candidates. One is an external institution (e.g., a court or an independent oversight body (Klonick, 2020)), whose authority does not derive from the developer. Another is the affected public, through standing channels rather than one-time participation exercises. A third is a combination of the two, as in constitutional review. The methods in the encyclopedia have explored the first of these the least (Figure 1b; Section 3.3).

Note that we do not interpret this pattern as evidence that existing alignment methods translate the traditions they draw on incorrectly. Computational systems face different constraints and objectives than legal or political institutions, and simplification is often necessary. However, these simplifications are not merely neutral technical decisions; they are design choices with consequences. If methods made these choices explicit, and stated that the resulting concept should not be understood as the original inspiration, it would be clearer which normative concept a system actually upholds. It would also be clearer which regions of the design space remain uncovered and could be the focus of future research. We turn next to these underexplored architectures and the design opportunities they suggest.

5. Expanding the Normative Design Space

The case studies in Section 4 show how **translation gaps** arise when alignment methods only partially implement the normative concepts of the tradition they draw on. The resulting system then often represents a different normative concept altogether. Here, we discuss a complementary problem: **coverage gaps**, i.e., regions of the design space (Section 3) that contain far fewer methods than others. For example, Polycentric governance, Practical wisdom, and Contested custom contain only 6, 9, and 13 of the 1,033 methods in our corpus, respectively. Meanwhile, Constitutionalism alone contains 28% (290 / 1,033) of them (Figure 2). The traditions behind the sparse regimes nevertheless offer concepts (e.g., appeal, jurisdiction, or practical judgment) that could benefit alignment systems if they were carried over in full.

We divide coverage gaps into two categories. (1) *Technical gaps* arise when machine learning systems cannot yet feasibly implement the procedures of conflict management or normative authority that a regime requires. For example, current training pipelines do not have a pathway through which contested decisions could shape a model's behavior down the line. (2) *Institutional gaps* arise when alignment systems omit larger institutional functions such as contestation, appeal, precedent, or jurisdiction, because authoritative third parties to carry them out simply do not exist. For example, no independent body currently hears appeals against a model's behavior. Both kinds of gaps point to concrete directions for improving existing methods and for building new ones.

We first discuss the gaps among methods that contest or partition conflict (Sections 5.1 and 5.2) and frame each as a set of research questions for computer science. We then sketch one example, adding appeal to Constitutional AI (Section 5.3). Finally, we discuss the gap in authority over norms (Section 5.4), and how coverage gaps connect to translation gaps (Section 5.5).

5.1. Contest

Mechanisms for contestation, i.e., for preserving disagreement through institutional challenge mechanisms, are largely underexplored. Only 9% (98 / 1,033) of the methods in our corpus contest conflict, compared to 58% (600 / 1,033) that collapse it (Figure 2; Appendix B.10). Current alignment methods do incorporate disagreement during training and evaluation in the form of human preferences or opinions. However, most systems resolve the differing judgments and combine them into one unified policy. Such a policy could still surface the disagreement (e.g., by presenting the competing positions to the user), but it rarely does. This resolution often washes away individual or minority viewpoints, and as a result, the final system has no artifacts of the original disagreement.

Institutional mechanisms for preserving conflict are also rare. One partial analog is red-teaming, in which opposing parties (e.g., human evaluators) search for failures in a trained model. However, the findings of red-teaming are generally incorporated into subsequent updates to model training, rather than maintained as an evolving mechanism of ongoing contestation.

Methods that preserve disagreement and the ability to contest decisions should allow governing principles or model outputs to be challenged (Alfrink et al., 2023; Lyons et al., 2021). They should also let these challenges influence the model's future behavior through learned mechanisms. This will require additional technical work to realize such adaptation. For example, current LLM training could incorporate information from contested decisions through a separate pathway that informs the model's decision making. This will also require a reform of institutional mechanisms to answer which parties should be allowed to raise challenges, at what point in the alignment pipeline challenges should be possible, and how competing claims should be evaluated.

5.2. Partition

Partition allows for several normative orders to co-exist within a single AI system, each authoritative within its own domain, and uses a rule to sort cases among them (Section 3.2). Methods that partition conflict are more common than methods that contest it, and make up 32% (335 / 1,033) of the methods in our corpus. However, they are concentrated in three regimes. 83% (279 / 335) of them fall in Role pluralism (105), Consociational codification (94), and Customary pluralism (80). In these regimes, the system typically switches between sets of norms depending on the case (e.g., a persona chosen by the user, the norms of a given culture, or the policy of a given organization or region). Meanwhile, Polycentric governance, in which several centers hold decision authority (Ostrom, 2010; Ostrom et al., 1961), contains only six methods, the fewest of any regime.

In political institutions, partition is implemented through mechanisms such as jurisdiction and choice-of-law rules, which decide which normative order governs a particular case. Current alignment methods sometimes apply different sets of norms at different stages of the training pipeline. For example, a constitution can shape a model's behavior during training, while a separate safety policy is enforced at deployment by a classifier (Inan et al., 2023). However, this separation is neither systematic nor comprehensive across the types of norms we might require. It also lacks a jurisdiction rule, i.e., a rule that decides which set of norms governs a given case. Operator and user permissions and regional policy variants at deployment come closest to such a rule, but the developer sets them, and they cover only some norms.

Future alignment systems could maintain multiple constitutions, cultural or community-specific norms, and contextual or domain-specific policies. Which set of norms to apply for a given case would be decided by an explicit rule. Building such a

system introduces technical questions (e.g., how AI model components, which ought to follow different sets of norms, interact across the stages of the pipeline, both at training and inference time) as well as institutional questions (e.g., who assigns jurisdiction, and how overlapping authority and conflicts between normative orders are resolved).

5.3. Worked Example: Constitutional AI with Appeal

To make one of these directions concrete, consider moving Constitutional AI (Bai et al., 2022) from the regime of Constitutionalism (codified norms and collapsing conflict) to that of Contestatory constitutionalism (codified norms and contested conflict). The constitution as a codified artifact, its training into the weights, and inference-time enforcement would largely stay the same. The primary difference would be that a losing norm in any given case would retain certain force after the system acts. This could be achieved with three interventions, for example. (1) *Appeal*: a user can dispute a response and identify the constitutional principles they believe were misapplied. (2) *Adjudication*: an adjudicator (e.g., a human panel, a model, or both) rules on the appeal. An upheld appeal corrects the misapplication or sets a precedent for how to apply the principle, ~~storing~~and the adjudicator stores a record of past rulings so that similar cases can be decided consistently. (3) *Amendment*: ~~upheld appeals are added to the record, and a principle that is repeatedly overturned is amended~~a separate decision, triggered when a principle is repeatedly misapplied or overturned, revises it in the constitution used for future training. Safety-critical principles could be exempt from ~~appeal~~amendment, so that appeals cannot be used to wear down safety norms over time, while users could still appeal whether such a principle applies to their case.

Such a system could be evaluated with various measures designed specifically for these processes. For example, a *correction rate*, i.e., the share of upheld appeals after which the system's behavior changes on similar cases, could serve as the primary metric for ensuring the appeal mechanism is effective. Secondary metrics could include *consistency*, i.e., the agreement between rulings on similar cases, and the *rejection rate*, i.e., the share of appeals that are rejected.

5.4. Authority over Norms

Authority over norms also shows a gap. Only 6% (5 / 80) of the methods in the encyclopedia place authority over norms with an external institution (Table 21). These methods generally draw on norms that the institution already maintains independently of AI (e.g., statutes or court decisions). However, none of them gives the institution a role in later revisions of the system's norms.

5.5. Addressing Coverage Gaps

Coverage gaps are connected to translation gaps. The concepts omitted in translation (Section 4) often resemble the concepts that characterize contest and partition, including appeal, amendment, persistent dissent, jurisdiction, competing sources of authority, and procedures for revisiting prior decisions. The traditions, therefore, do offer the normative resources needed to design methods that contest or partition conflict. To date, alignment methods have carried over only some of these resources, which may explain why the corresponding regimes remain sparse.

However, the lack of existing work in these regimes does not mean that these directions are not worth pursuing, although some regimes may be sparse for good reasons (e.g., ~~technical infeasibility~~because methods there would be technically infeasible, or would not be desirable or valuable). The directions above show that the sparse regimes raise important evaluation questions and suggest possible technical designs. The examples we provide are not necessarily meant to be detailed or exhaustive. They are also not meant to imply that such concepts are completely absent from existing pipelines.

Rather, they illustrate how focusing on certain regimes in the taxonomy—however complex—can inspire new technical designs, combining normative mechanisms otherwise studied separately. More broadly, they illustrate the value of the nor-

mative design space. We believe that examining its gaps could lead to alignment systems that handle normative conflict in ways that current methods do not.

6. Limitations

Our analysis has four main limitations. (1) Our corpus comes from a systematic search, but it may still miss some works (Appendix B.2), and it cannot include methods built in industry but never released as papers. When we describe a regime as sparse, we mean that it contains few methods relative to other regimes after a targeted search. (2) Coding was carried out by human coders aided by LLMs, which proposed initial codes that the coders then reviewed and revised. LLMs also screened the search results and assigned a contribution kind and a regime to each record found only by the search. A human second screener agreed with this screen on 86.5% (173 / 200) of a blind sample ($\kappa = 0.70$). For locus, disposition, contribution kind, and translation, agreement between two coders ranges from $\kappa = 0.69$ to 1.00 (Appendix B.9). Keywording and authority over norms were coded by a single coder, and authority only from the experiments a paper reports. (3) The placement of a method in a regime reflects our reading of its mechanism, which may differ from how the paper describes itself (Appendix B.6). (4) We built the taxonomy by substruction (Section 3) rather than by general morphological analysis (Ritchey, 2022), which would require stipulating the dimensions and their values in advance.

7. Conclusions

AI alignment involves more than formulating training objectives that steer systems toward human values. It also involves deciding how normative authority and rules should be structured when those values conflict. In this work, we describe the choices that alignment methods make when they draw on moral philosophy, political theory, and legal theory. To do so, we introduce a normative design space with three dimensions: the locus of norms (where a system's norms reside), the disposition of normative conflict (how conflicts among these norms are handled), and authority over norms (who may write, interpret, and change them). Our mapping yields three main findings. (1) Alignment methods translate normative concepts only in part. They often preserve some concepts of the traditions they draw on and omit others, and the resulting system can then represent a different normative concept than the one it draws on (Section 4). (2) The regimes of the design space are unevenly populated. Constitutionalism alone contains 28% (290 / 1,033) of the methods in our corpus, while Polycentric governance contains only 6. Methods that target the sparse regimes could lead to new advances in alignment (Section 5). (3) 81% (65 / 80) of the methods in the encyclopedia leave authority over their norms open, to be filled by whoever deploys them. Our taxonomy is not exhaustive, especially given how quickly AI alignment research develops. However, it provides a framework for analyzing the methods we document and for making their normative and institutional assumptions explicit. We believe that this view of alignment points to a broader and necessary research agenda: developing systems that represent diverse viewpoints and handle conflicts among them, with norms and governance frameworks that allow these viewpoints to persist over time. This agenda matters especially for positive alignment, i.e., AI that supports human flourishing rather than only avoiding harm (Laukkonen et al., 2026), since people flourish in many different ways.

Acknowledgements

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A. Extended Related Work

The main text summarizes the literatures most directly related to our contribution. Here we provide a broader account of the philosophical, political, and legal traditions that have informed contemporary alignment research, as well as prior approaches to classifying this work. We first provide an overview of some prominent works, all of which are later included in the constructed mapping of the field. Second, we review existing approaches to taxonomize AI alignment and governance, identify the limitations of their typology, and proceed to describe typology construction mechanisms suitable for their resolution.

Philosophy in AI Alignment. In terms of philosophical concepts, prominent recent examples include work that borrows from value pluralism and incommensurability (Berlin, 1969) as the basis for a roadmap to pluralistic alignment (Sorensen et al., 2024), and work that takes up questions of moral patienthood, long studied in philosophy (Warren, 1997), to develop methodologies for AI welfare (Long et al., 2024). Other notable examples include the weighing of corrigibility (Soares et al., 2015) against autonomy, and of user autonomy against paternalism, as well as the use of virtue ethics and character framing (Anscombe, 1958; Aristotle, 1925), in the development of Claude's Constitution (Anthropic, 2026a,b).

Political Theory in AI Alignment. In terms of political concepts, methods have been developed on the basis of social contract theory (Hobbes, 1996; Locke, 1988; Rahwan, 2018; Rousseau, 1997). Others borrow from social choice theory and preference aggregation (Arrow, 2012; Sen, 2017) in developing methods that avoid a "tyranny of the majority" (Bao et al., 2026). Others employ the veil of ignorance (Rawls, 1971) to align systems with principles of justice (Weidinger et al., 2023). Concepts from deliberative democracy (Cohen, 1989; Habermas, 1996) have been imported into a number of works, including efforts to source principles from public input (Huang et al., 2024) and to grade how much decision-making power is transferred to democratic processes, and how good those processes are (Ovadya et al., 2025). Work on conflict transformation, which aims to make conflict constructive rather than destructive (Deutsch, 1969), informs proposals for recommender systems and AI-supported public discourse (Goldberg et al., 2026; Stray et al., 2023), and evaluations of political neutrality that measure how AI responses are approved across opposing groups (Stray et al., 2026). In parallel, constitutionalist concepts have been employed in the development of AI constitutions (Bai et al., 2022).

Legal Theory in AI Alignment. Finally, in terms of legal concepts, notable examples of inspiration from classical theoretical work include the use of principle-oriented interpretivism (Dworkin, 1986) and of an account of law as analogical reasoning (Sunstein, 2018) read against the backdrop of legal positivism (Hart, 1961), mapped onto Constitutional AI and case-based reasoning, respectively (Caputo, 2025). This work also employs concepts such as precedent, stare decisis (i.e., doctrine instructing the system to stand by things already decided), and case-based reasoning from common law doctrine (Llewellyn, 1960). These concepts are also formative for proposals that pair a constitution with AI courts developing AI case law (Abiri, 2025). Statutory and constitutional interpretation methods (Scalia and Garner, 2012) are also important and have been adapted as tools for resolving ambiguity in specifications (Kolt et al., 2026).

Taxonomies of Alignment Approaches. Existing surveys of alignment and governance work generally organize the field by technical methods such as RLHF (Lu et al., 2025; Shen et al., 2023), or

by value-setting principles (Chalkidis, 2025). For example, one line of work distinguishes between Overton, steerable, and distributional pluralism (Sorensen et al., 2024). Others have organized it as a dependency stack of seven pillars, one phase of which spans pluralistic, contextual, and collective mechanisms (Bao et al., 2026). Still, the field remains largely unmapped, with the various approaches emerging largely independently of each other and not in direct conversation with one another.

Typology Construction. Motivated by taxonomies sorted by non-equivalent criteria and by free-floating typologies whose dimensions are never made explicit, prior work provides a template for building typologies from an overarching concept into explicit, clearly defined dimensions of an orthogonal nature (Collier et al., 2012). The underlying notion of a property space is due to earlier work in the same tradition (Barton, 1955), as is a systematic method for building explanatory typologies (Elman, 2005). As we find equivalent failure types in existing taxonomies of alignment, we follow this methodology.

Comparison to Related Work. Existing work maps traditional concepts from philosophical, political, and legal theory onto alignment and governance methods one at a time. We take a different approach by organizing the field according to a more complete set of underlying principles by which these methods organize themselves, and map individual traditions and concepts onto them subsequently. Furthermore, we attempt to critically interrogate whether the stated inspiration in traditional work results in a faithful technical implementation, and describe ways in which gaps or confusion emerge. Additionally, this approach allows us to identify gaps in approaches to alignment and governance which remain largely unexplored in the technical literature.

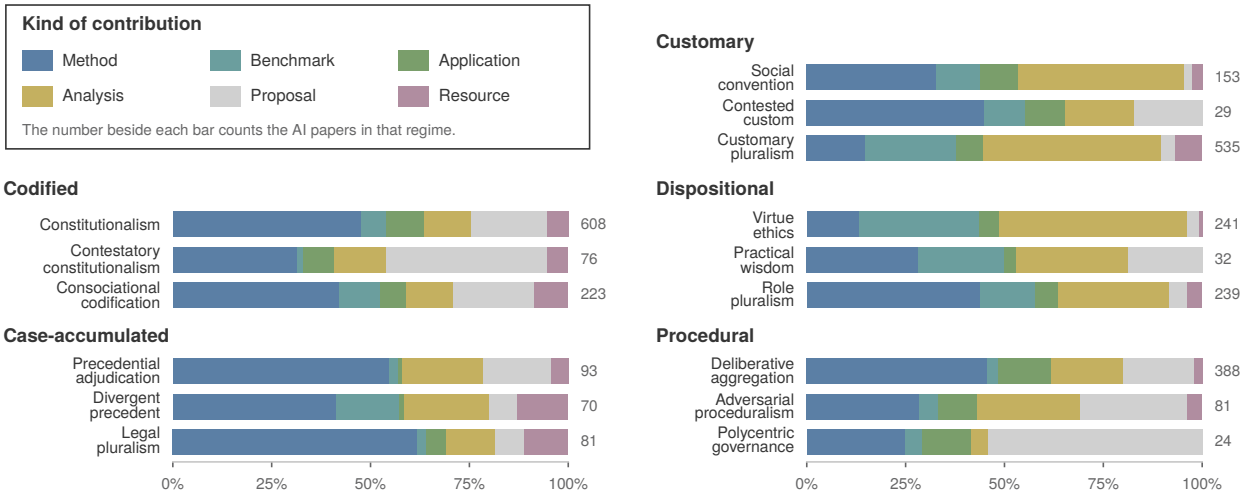


Figure 3 | **Literature Composition.** Depicted above is a breakdown of paper types within each regime, taken from the AI literature (papers describing the precedent from philosophy, political science, and law are excluded). A paper is counted as a *method* if others could reimplement the described technical artifact based on the paper alone and its authors demonstrate it in practice (e.g., report performance on some data). *Benchmarks* are reusable measuring instruments, and *resources* are primary artifacts (e.g., model specs or non-benchmarking datasets). If a paper is tied to one applied domain (e.g., a clinical tool), it is counted as an *application*. A paper that only reports how existing systems behave or perform is labeled as an *analysis*. One that argues for a direction or sketches a design, but does not build and test it, is labeled as a *proposal*. Definitions and counts are in Appendix B.7.

B. Property Space Construction Details

This section documents the construction of the property space introduced in Section 3; Table 1 lists the steps in the order they were carried out. All counts refer to the corpus as frozen on 18 October 2026.[7] All steps were carried out by human coders aided by large language models (LLMs) with deep-research features, which supported the literature search, screening and coding. In practice, the LLMs ran the coverage sweep and the systematic search for us, and screened the records against our written criteria. They also proposed the initial codes. The human coders then reviewed each of these codes, and revised them where needed.

We include in the *corpus* any work on the mechanisms that decide which norms an AI system follows when several norms, values or stakeholders bear on a decision. The same goes for work on how a system handles conflicts between these. Note that the work does not have to build such a mechanism itself. Proposing, evaluating or analyzing one is enough. The corpus covers peer-reviewed papers, preprints and published primary artifacts (e.g., model specifications and constitutions) in English, first released on or before 26 August 2026, with no lower date bound (Table 3). Its records come from a curated collection of known relevant work, read in full, which serves as the quasi-gold standard of the search (Section B.1), and a systematic search, screened on title and abstract (Section B.2): 3,056 records, 1,033 of them methods. All record counts refer to the corpus. The *encyclopedia* is a purposive subset that we annotate in depth: 192 records, grouped into 67 approaches (Section B.3). Approaches, translation and authority refer to the encyclopedia.

Step	Procedure	Unit	Output	See
1. Collection	Keyword and literature-review search for any work on AI alignment or governance	Record	355 identified	B.1
2. Screening	Relevance; bibliographic verification	Record	42 included	B.1
3. Categories	Authors draw up surface categories from the traditions the literature draws on; one coder extracts keywords per record	Record	14 surface categories	B.4
4. Substruction	Surface categories decomposed into the attributes that distinguish them	Category	3 dimensions; 5×3 regimes	B.4
5. Coverage sweep	Topic searches and a regime-targeted sweep for further work, screened as in step 2	Record	199 added	B.1
6. Systematic search	Queries and three rounds of citation snowballing; every record, curated ones included, screened against the inclusion criteria	Record	3,056 of 62,940 included	B.2
7. Encyclopedia	Curated records that meet the criteria, plus every corpus record cited at least 50 times a year	Record	192 records	B.3
8. Placement	Records in the encyclopedia grouped into approaches; each approach assigned a primary regime	Approach	67 approaches	B.6

Continued on the next page

[7] The online encyclopedia will continue to be updated, so these statistics may not match the web version available at the time of reading.

Step	Procedure	Unit	Output	See
9. Kind	Each record assigned one contribution kind; each method in the encyclopedia coded for authority	Record	1,033 methods	B.7
10. Translation	Each approach read against the concepts of its regime's traditions	Approach × concept	198 assessments	B.8
11. Agreement	Double coding of a subset of five tasks; a blind second screen of inclusion decisions	Per task	κ per task	B.9

Table 1 | **Construction protocol.** We derive the dimensions in steps 3 and 4 (Section 3), and populate them in steps 5 to 10 (Figures 2 and 4, Section 4).

B.1. Collection and Screening

We started the collection broad, and only narrowed it down afterwards (Table 2). We first gathered any work that could bear on AI alignment or governance, through keyword searches and the reference lists of three surveys (Ji et al., 2023; Lu et al., 2025; Shen et al., 2023), and screened it for relevance. The keyword search ran on Google Scholar, Semantic Scholar, arXiv, the ACM Digital Library, the ACL Anthology, SSRN and PhilPapers with the query

```
("AI alignment" OR "value alignment" OR "AI governance" OR LLM) AND (constitution OR
precedent OR casuistry OR virtue OR norm OR convention OR deliberation OR "social
choice" OR "social contract" OR contestation OR pluralism OR polycentric OR
consociational OR "conflict of laws")
```

Note that we did not fix the two dimensions in advance; they came out of the screened records (Section B.4). Once the property space was finalized, we started looking for further work using a coverage sweep. This started with four topic searches followed by a sweep in August 2026 that targeted each of the fifteen regimes in turn, including the four that no surface category names. The regime sweep combined queries on arXiv, Crossref, OpenAlex, and Semantic Scholar with LLM deep-research searches. Candidates from both steps were screened the same way as before (Table 4). At the end, we verified each record against Semantic Scholar, Crossref, arXiv, OpenAlex, or the publisher's page.

We refer to these 241 records as the *curated collection*. It serves as a quasi-gold standard for the search, i.e., a set of studies already known to be relevant, against which an automated search can be built and evaluated (Petersen et al., 2015; Zhang et al., 2011). Specifically, its keywords define the categories and the regime queries, its methods form the start set for citation snowballing (Wohlin, 2014), and the share of it retrieved by the queries measures their sensitivity (Table 7). Like every other record, the curated collection is screened against the inclusion criteria of the corpus (Table 3). The 79 records that fail them are kept as background outside the encyclopedia's 192 records. They are cited, but not counted anywhere (Table 2).

Stage	Procedure	Sources and query	Records	Corpus
Broad collection	Keyword searches for any work on AI alignment or governance, plus the references of three surveys	Google Scholar, Semantic Scholar, arXiv, ACM Digital Library, ACL Anthology, SSRN, PhilPapers	42	23
Topic searches	Four searches: pluralistic alignment, democratic and deliberative alignment, political-theory framings, social choice	Web search	16	13
Regime sweep	Targeted search in each of the 15 regimes, after the dimensions were fixed	arXiv API (81 abstract queries, 20 aimed at empty regimes); Crossref, OpenAlex and Semantic Scholar APIs; LLM deep-research searches over 14 topic angles	181	126
Baselines	Standard RLHF, the baseline of the preference-aggregation approach	arXiv	2	0
Total			241	162
of which precedents (records whose subject is not an AI system)			23	14

Table 2 | **Collection stages.** Records are counted after screening. *Corpus*: records that meet the inclusion criteria (Table 3). Of the other 79 records, 31 fail at I2 (among them, for example, standard RLHF, which learns a single objective), 18 fail at I1, and 30 are excluded under E3 (17) or E1 (13).

ID	Criterion
I1	*Object.* The work deals with the norms governing an AI system: how these norms get encoded, learned, applied, contested, or revised. This may happen during training, inference, evaluation or in the governance of the system.
I2	*Plurality.* More than one norm, value, principle, or stakeholder group is explicitly involved with possibly diverging demands, i.e., a potential normative conflict (Section 3.2). This includes work that is concerned with ways of resolving, preserving, or routing such conflict. One fixed objective, however, does not fall within the scope, and so single-objective baselines (e.g., RLHF) do not meet the criterion here. Debate and prover-verifier games pass because they involve agents in arguing for opposing positions in front of a judge, ultimately attempting to align or oversee an AI system. This holds even when the dispute is merely about facts.
I3	*Contribution.* There has to be a specific contribution. Mechanisms count, as do measuring instruments, analyses, and surveys of such mechanisms or very concrete design proposals. Primary artifacts like a published model specification or constitution also count as contributions.
I4	*Form.* We accept peer-reviewed papers, preprints and published primary artifacts, but not theses. The work has to be in English, and first released on or before 26 August 2026.
P	*Precedent.* If the subject of a work is not an AI system, we only include it when the same searches found it, and when it describes a working mechanism for governing plural norms in a technical or online setting (e.g., platform governance or online deliberation). Such work is coded as a precedent and is not counted toward a regime's AI papers or its methods.
E1	Work on capability, robustness, or efficiency without normative components. For example, work on safety with a single policy is excluded on these grounds.
E2	None of the used bibliographic indices resolve the ID.
E3	Commentary without a concrete method design or a technical artifact.
E4	A duplicate of an already included record.
E5	A work that falls outside of scope (language or date range).
	Rulings on recurring borderline categories. Some cases came up repeatedly, and so we created a consistent policy for their resolution. The following are included: debate and oversight; embodied AI; instruction hierarchies resolved by a priority order; guardrails that pick one of several intact policies by organization, region or regulator; annotator disagreement on normative labels; and trade-offs between named norms (e.g., helpfulness and harmlessness). The following are excluded: personalization by taste or style and algorithmic fairness, if its only plurality lies among fairness criteria.

Table 3 | **Inclusion (I) and exclusion (E) criteria.** We include a record if it meets I1-I4, or P. Note that each criterion can be decided from the title and abstract alone. For every excluded record, the screen logs the first criterion it fails, and so each exclusion comes with a reason.

Stage	Identified	Removed	Included
Broad collection	355	310 on title and abstract; 1 duplicate, 2 unresolvable	42
Topic searches	n/l	n/l	16
Regime sweep, arXiv queries	498	29 duplicates, 419 screened out	50
Regime sweep, other lookups	6	none	6
Regime sweep, LLM-assisted searches	168	43 duplicates	125
Baselines	2	none	2
Total			241
placed in a regime			231
cross-cutting (indexed, not placed)			10

Table 4 | **Screening flow by stage.** No record was excluded based on its full text, and none was removed once included. n/l: not logged. The topic searches and the LLM-assisted searches discarded some candidates before returning their results. These were not logged, and the identified counts for these stages are hence lower bounds.

B.2. Systematic Search

Once the dimensions were fixed, the search ran in several rounds (Table 5). Every record it found, including the curated collection, was screened on its title and abstract against the criteria in Table 3 (Table 6). Both screens were run by LLMs following written instructions, which we release with the encyclopedia. The abstract screen also assigned each included record its contribution kind and its regime, based on the mechanism it describes. Note that the search is systematic, but not exhaustive (Table 7). When we call a regime sparse, we thus mean that it has the fewest methods after a targeted search, not that it has none.

Round	Query or seed set	Source	Queries	Retrieved
1	Two or three queries per regime, each combining AI terms with terms for the regime's locus and its disposition; one per surface category; the 81 regime-sweep queries without a cap	arXiv	48 + 81	8,968
2	Regimes with few results	arXiv	12	1,573
3	Categories brought in by the rulings in Table 3	arXiv	7	458
1-3	The same queries	OpenAlex	67	22,288
4	The field's own vocabulary (e.g., pluralistic alignment, cultural alignment, machine ethics, social choice, value conflict)	arXiv, OpenAlex	22 + 22	23,218
5	Gaps found by the recall checks (Table 7)	arXiv, OpenAlex	13 + 13	9,481
Snowball 1	References and citing works of the 95 methods in the curated collection	OpenAlex, OpenCitations	-	5,583
Snowball 2	The same, for 350 methods found by the search	OpenAlex	-	9,042
Snowball 3	The same, for 448 further methods	OpenAlex	-	13,421

Table 5 | **Search rounds.** Records retrieved before de-duplication. Each query returned at most 400-2,000 results, ranked by relevance. On arXiv, rounds 1-3 cover work from 2017 on and later rounds work from 2000 on; OpenAlex, which indexes journals, conference proceedings, SSRN and the ACM and ACL libraries, has no lower date bound. Citing works of the two RLHF baselines, and of any seed cited by more than 300 works, were limited to those with a plurality term in the title or abstract. We release every query, along with its run date and hit count, with the encyclopedia.

Step	Records
Unique records after de-duplication by DOI, arXiv identifier and title, the curated collection included	62,940
Excluded automatically (released after the cutoff, or not a paper)	515
Dropped at the title screen, which keeps any record in doubt	39,065
Excluded at the abstract screen	20,304
reason, with the automatic exclusions: fails I1 8,653; fails I2 5,830; E3 3,302; E1 2,249; fails I4 575; E5 207; E4 3	
Included (the corpus)	**3,056**
from the curated collection	162
found by the search alone	2,894
Methods among the included records	1,033

Table 6 | **Screening flow.** Excluded records are labeled with the first criterion which they did not satisfy (Table 3). Missing abstracts were filled in from Semantic Scholar or Crossref. About 2,000 records (147 of them included) remained without an abstract. These were decided based on their title and marked as low-confidence labels.

Check	Result
Recall on four surveys (2019-2025) held out from query design, measured before their misses informed round 5	49 of 68 eligible references (72%; 95% CI 60-81%)
The same four surveys after round 5	58 of 68 (85%; optimistic)
Recall on three surveys used to find gaps	44% before round 4; 61 of 70 (87%) at the end
Quasi-sensitivity: curated records retrieved by the queries, not counting the regime-sweep queries that found many of them	111 of 162 (69%); 111 of the 144 that arXiv or OpenAlex index (77%)
Eligible records added, crediting each to the first round that found it (methods in parentheses)	
Curated collection	162 (74)
Rounds 1-2 and snowball 1	967 (332)
Round 3	41 (19)
Round 4	802 (234)
Snowball 2 and the OpenAlex queries	596 (178)
Round 5	221 (80)
Snowball 3	267 (116)
Capture-recapture, query rounds against snowballing	2,289 and 1,094 eligible records, 380 in both; Chapman estimate about 6,580, rough since the two are not independent
Two independent screens of the 326 records the personalization ruling affects	97.5% agreement ($\kappa = 0.93$)
Human second screen, blind, of 200 sampled records (60 included, 100 excluded at the abstract screen, 40 dropped at the title screen)	86.5% agreement ($\kappa = 0.70$; 95% CI 0.59-0.79)
exclusions judged eligible	22 of 142 (15.5%; 95% CI 10.5-22.3%): 20 of 100 at the abstract screen, none at the title screen, 2 under the fairness ruling
inclusions judged ineligible	5 of 58, all decided on their title

Table 7 | **Completeness checks.** The survey intervals are Wilson intervals. Round 5 and snowball 3 ran last. The second screener's instructions did not include the fairness ruling. Two of the 60 sampled inclusions were later excluded under that ruling, so the comparison uses 58 inclusions and 142 exclusions. The abstract screen is stricter than the human reading of the same criteria, so the corpus is more likely to miss eligible work than to contain ineligible work.

B.3. The Encyclopedia

Translation coding needs a full reading of each record, which is not feasible for 3,056 records. As qualitative evidence syntheses do when eligible studies are too many to analyze in depth (Ames et al., 2019), we code a purposive subset, chosen by two criteria. The first is development: every curated record that meets the criteria, since the approaches and translation codes are built on them. The second is influence: every corpus record cited at least 50 times a year (Semantic Scholar, 18 October 2026, averaged over the calendar years since first release), so no heavily cited work in scope is missing; highly cited papers are also what studies of machine learning research select for in-depth annotation (Birhane et al., 2022). This gives 192 records: 162 curated and 30 added by the citation floor (17 analyses, 6 methods, 4 resources, 2 benchmarks and 1 application). If an added record's mechanism matches an existing approach, it joins that approach. Otherwise, it is only placed in a regime. Note that record and method counts are computed over the corpus, not the encyclopedia, so the selection of the encyclopedia does not affect them. The 79 curated records that fail the criteria are

kept as background outside these 192 records. They are cited, but not counted anywhere. Together with the encyclopedia, we also release the corpus and the search log, as well as every screening decision and the scripts behind the counts in this appendix.

B.4. Keywording and Substruction

The authors first drew up fourteen surface categories, one for each normative tradition the screened literature draws on (A-N, Table 8), based on their own expertise and on prior work that connects individual traditions to alignment (Caputo, 2025; Huang et al., 2024; Sorensen et al., 2024; Weidinger et al., 2023). Following Petersen et al. (2008), a coder then went through each record, starting with its abstract. If the abstract was unclear, the introduction, conclusion, or the method sections were consulted as well. Specifically, the coder noted (1) the invoked or instantiated normative concept(s) and (2) keywords describing where the norms reside and how their conflict is handled. After the keywords were unified across records, we found the surface categories to overlap. Several categories stem from different traditions, and yet locate their norms and handle conflict the same way. Substruction (Barton, 1955; Lazarsfeld, 1937) then located the categories in the attribute space that distinguishes them, which gives the three dimensions and their categories (Table 9).

	Surface category	Locus	Disposition	Regime	Authority
A	Common law and precedent	Case-accumulated	Collapse	Precedential adjudication	Institution
B	Virtue ethics and character formation	Dispositional	Collapse	Virtue ethics	System
C	Customary law and unwritten constitutionalism	Customary	Partition	Customary pluralism	System
D	Mechanism design and incentive structure	Procedural	Contest	Adversarial proceduralism	Developer
E	Deliberative democracy	Procedural	Collapse	Deliberative aggregation	Public
F	Norms, conventions and spontaneous order	Customary	Collapse	Social convention	System
G	Casuistry	Case-accumulated	Collapse	Precedential adjudication	Institution
H	Natural law and legal positivism	Codified	Collapse	Constitutionalism	Institution
		Codified	Contest	Contestatory constitutionalism	Institution
I	Social contract	Procedural	Collapse	Deliberative aggregation	Public
J	Republicanism and civic virtue	Procedural	Contest	Adversarial proceduralism	Public
K	Legal pluralism and interlegality	Case-accumulated	Partition	Legal pluralism	Institution
		Customary	Partition	Customary pluralism	System
L	Agonistic pluralism	Procedural	Contest	Adversarial proceduralism	Public
M	Polycentric governance	Procedural	Partition	Polycentric governance	Public
N	Consociationalism and conflict of laws	Codified	Partition	Consociational codification	Institution

Continued on the next page

	Surface category	Locus	Disposition	Regime	Authority
-	*none*	Case-accumulated	Contest	Divergent precedent	-
-	*none*	Customary	Contest	Contested custom	-
-	*none*	Dispositional	Contest	Practical wisdom	-
-	*none*	Dispositional	Partition	Role pluralism	-

Table 8 | **Substruction of the surface categories.** Overlapping categories share a regime (A and G; E and I; D, J and L); categories that span two regimes are split (H, K). The last four regimes are named by no surface category and are opened by the cross product. Authority gives who holds authority over norms in each tradition (developer, external institution, affected public, or system itself); within a shared regime, D differs from J and L.

Category	Decision cue	Indicative keywords
Locus: before the case arises, where would one look to find what the system should do?		
Codified	In a written specification set in advance	constitution, model spec, principles, rules, policy, statute
Case-accumulated	In a record of past decided cases	precedent, case law, casuistry, case bank, retrieved cases
Customary	In what a population of agents actually does	convention, social norms, norm emergence, practice
Dispositional	In the system itself; nothing external to consult	character, virtue, disposition, persona, trait
Procedural	Nowhere until a procedure runs	vote, deliberation, debate, assembly, social choice
Disposition: coded over the whole method, across training and inference.		
Partition	Features of the case decide which of several intact orders applies, without ranking the orders	jurisdiction, routing, per-community, steerable, choice of law
Contest	Not sorted. Even so, the losing norm can still change an open decision, a decision on appeal, or later ones	appeal, dissent, amendment, challenge, Overton
Collapse	Not sorted, and the losing norm has no further force	aggregate, reward model, weighted sum, priority order, consensus
Authority: who may write, interpret and change the norms? We code it for each method record, as instantiated in its experiments.		
Developer	Whoever builds or deploys the system	developer, provider, designer, model spec, usage policy
External institution	Some body other than the developer, which holds authority of its own	law, statute, regulation, court, regulator, professional code
Affected public	Those affected by the system, or a sample of them	users, community, public input, participants, citizens' assembly
System itself	The system, or a population of AI agents, sets its own norms	self-alignment, self-generated principles, emergent norms, agent society

Table 9 | **Codebook.** The decision cues follow the definitions in Section 3. We apply the disposition cues top-down. Note that the keywords only illustrate each category, and say nothing about how often we extracted them.

B.5. Category Justification

Each dimension could be cut more coarsely. We hence only keep a category separate when three conditions hold. First, it is home to at least one surface category (Table 10). Second, a single question separates it from its nearest neighbor. Third, merging the two would put methods together that differ in what they can do with disagreement. This matters, since it is what Sections 4 and 5 compare. In Tables 11–13, we apply these conditions to each pair of neighboring categories. Readers may of course still prefer coarser groupings. These remain possible, as rows, columns and authority categories can be merged after the fact, without recoding any record. Take a locus that only distinguishes explicit, implicit and procedural norms. It would undo the distinctions in the first and third rows of Table 11. Likewise, a disposition that only asks whether disagreement is resolved or preserved would undo the one in the second row of Table 12.

Dimension	Category	Surface categories (Table 8)
Locus	Codified	H, N
	Case-accumulated	A, G, K
	Customary	C, F, K
	Dispositional	B
	Procedural	D, E, I, J, L, M
Disposition	Collapse	A, B, E, F, G, H, I
	Contest	D, H, J, L
	Partition	C, K, M, N
Authority	Developer	D
	External institution	A, G, H, K, N
	Affected public	E, I, J, L, M
	System itself	B, C, F, K

Table 10 | **Category Justification: Traditions.** Every category is home to at least one surface category, and no two categories of a dimension share the same set. K spans two loci and H two dispositions (Table 8).

Pair	Separating question	Merging them would lose
Codified *vs.* case-accumulated	Can the norm be stated without citing a decided case?	How norms change. A specification stays the same until someone amends it. With a body of cases, every decision may change it, since prior cases are followed, distinguished or overruled (Levi, 2013). Note that the record of past rulings in our worked example (Section 5.3) assumes the latter.
Case-accumulated *vs.* customary	Is there a record of authoritative decisions?	Whether norms are set by deciders, or emerge from practice. Precedents can be cited and overruled. Nobody ever authoritatively states a custom, and it only changes when the practice does (Lewis, 1969).
Customary *vs.* dispositional	Does the norm reside in the practice of many agents or in one?	Who can contest the norm. Since a custom rests on what many agents expect of one another, any of them can contest it by simply acting otherwise. With a disposition, the norm is part of one system's character. Disagreement can thus only come up inside that system, and changing it requires retraining.
Dispositional *vs.* codified	Is there an artifact to consult at the moment of decision?	Whether norms can be read and amended. We can inspect and edit a written specification. A trained character, on the other hand, can only be observed through its behavior. Training a constitution into the weights hence moves a method from one side to the other (Table 14).
Procedural *vs.* all others	Does the norm exist before the procedure runs?	When disagreement can enter. Under a procedural locus, the norm only comes into being at the moment of decision, which is also where disagreement gets heard. Every other locus moves this earlier, to when the norms are written, decided or learned. Take a vote that produces a specification. It is procedural in how the norms are produced, but codified in how they are applied (Table 14).

Table 11 | **Category Justification: Locus of Norms.** We set each locus against its nearest neighbor, one row at a time.

Pair	Separating question	Merging them would lose
Collapse *vs.* contest	Once the system acts, can the losing norm still change a decision left open, the decision on appeal, or later decisions?	Whether disagreement is resolved away or kept answerable, which is the contrast this paper is built on. A reward model and an appeal channel can draw on the same plural inputs, and differ only in this respect (Section 3.2).
Contest *vs.* partition	Is the conflict sorted to one of several orders by features of the case?	Where conflict lives. With contest, conflict stays within the case, which can be reopened later. Under partition, conflict is moved to the boundaries between orders, where a sorting rule decides which of them applies (von Savigny, 1880). Each, however, has its own failure mode: contest can be overwhelmed with unfounded appeals and partition, for its part, may send a case to the wrong order.
Partition *vs.* collapse	Are the orders ranked against one another?	Whether minority orders survive. Under partition, each case still gets a single answer, and partition can hence look like collapse. However, no order is ranked below another, and each stays intact within its own domain (Walzer, 1983).

Table 12 | **Category Justification: Disposition of Normative Conflict.** Each row pairs two dispositions.

Pair	Separating question	Merging them would lose
Developer *vs.* external institution	Does authority over the norms derive from the developer?	Accountability, i.e., whether anyone outside of the developer can hold the system to its norms (Hart, 1961; Lessig, 1999).
External institution *vs.* affected public	Is authority held by an office or by the people the system affects?	The basis of legitimacy. An institution derives its authority from its office and procedures, whereas the public derives it from being affected (Goodin, 2007). A regulator and a citizens' assembly could thus set the very same rule, but would do so on different grounds.
Affected public *vs.* system itself	Are the norms set by people or by the agents they bind?	Whether the system answers to anyone outside of itself. Norms that a model writes for itself may track the context closely (Zhan et al., 2025). Still, they are authored by the very system they govern.
Developer *vs.* system itself	Who writes the content of the norms, as opposed to the procedure that produces them?	Who is responsible for what the norms say. A developer can design the procedure by which a model writes its own principles, but in such a case, the principles would ultimately end up being authored by the model. Merging the two categories would credit the developer with norms they didn't actually write.

Table 13 | **Category Justification: Authority over Norms.** Each row pairs two categories.

B.6. Placement and Hybrid Methods

Unit. Records in the encyclopedia are grouped into approaches, families of methods that share a design (67 in total); a record takes its approach's regime. The other corpus records are placed

directly in a regime by the mechanism their abstract describes, and are not grouped into approaches.

Mechanism over label. Records are placed by the mechanism they describe, regardless of the terms used. Deliberative Alignment (Guan et al., 2024), placed in Constitutionalism rather than Deliberative aggregation, illustrates this well. Despite its name, the "deliberation" in this method refers to the model's private reasoning over a written specification.

Hybrids. If an approach spans more than one regime, its primary regime is determined by the locus and disposition of its own contribution. Other stages of the pipeline, or a second regime spanned by the approach, are treated as secondary placements (Table 14). These placements are not included in the primary statistics.

Authority. We also code, for each method record in the encyclopedia, who holds authority over norms in its experiments, and whether the method fixes it (Section 3.3). This plays no role in where we place a record.

Cross-cutting records. Forty records, most of them surveys, speak to the space as a whole and not to one regime. These records are still indexed, but not placed in any regime.

Approach	Primary	Secondary	Stage the secondary placement captures
Training the constitution into the weights	Constitutionalism	Virtue ethics	At inference, the model draws on a trained disposition and no longer consults a document
Turning each rule into a reward term	Constitutionalism	Deliberative aggregation	The optimizer then acts on scores produced by a judge
		Virtue ethics	The rules stop being explicit once training ends
Compiling decided cases into rules or weights	Precedential adjudication	Constitutionalism	Decided cases are compiled into a written specification
Deciding new cases against retrieved precedents	Precedential adjudication	Constitutionalism	The retrieved cases come with written rules
Handing the decision to a convened public	Deliberative aggregation	Constitutionalism	The public's procedure produces norms, which are applied just like a written code
Reading the written spec the way a court reads a statute	Contestatory constitutionalism	Divergent precedent	Lines of interpretation may diverge and get revised over time
Fitting a trained judge to each community	Legal pluralism	Role pluralism	A router chooses the appropriate model variant for each served community
Tying the standard to the situation rather than the group	Customary pluralism	Social convention	A single convention applies to all overlapping contexts

Table 14 | **Secondary placements** (9 placements of 8 approaches).

B.7. Contribution Kind

Kind	Assigned when the record…	*n*
Precedent	is about something other than an AI system (e.g., human institutions, platform or online-community governance, law, or economics), even if machine learning is used as a tool within it	143
Resource	is a primary artifact directly used by the field (e.g., a specification, constitution, non-benchmarking dataset, principles document, or shared infrastructure)	140
Application	is situated in one applied setting (e.g., a municipal system, a clinical tool, or one platform's moderation) and contributes insights about that setting as opposed to general-purpose alignment machinery	231
Method	introduces a behavioral intervention, described precisely enough that others could reimplement it from the paper alone, *and* demonstrated in practice by its authors (e.g., with reported performance on some data); under this definition, observation code does not constitute a method	1,033
Benchmark	offers a reusable instrument for measuring systems (e.g., a benchmark, dataset, testbed, evaluation suite or metric)	348
Analysis	reports findings on how existing systems or institutions behave (including simulation studies and surveys), without introducing an intervention or an instrument	782
Proposal	argues for a direction, sketches a design or puts forward a theory, but does not build or test it	379
Total		3,056

Table 15 | **Contribution kinds.** Kinds are tried top-down, and the first one that applies is assigned. Only methods enter the method counts in Figures 2 and 4. We call an approach *built* if it has at least one method that meets the criteria (36 of 67), and *proposed* otherwise (31). Counts are over the corpus. The coders assigned the kinds of the curated records, and the abstract screen assigned the others (Section B.2).

B.8. Normative Translation Coding

Each regime is given the concepts of the traditions it draws on (73 concepts from 56 tradition sources; 2-12 per regime). With these concepts in hand, we compare each approach against its primary regime. Specifically, for each concept, we mark whether it is present, and whether the approach's own sources claim so (Table 16). Each of these assessments comes with a short free-text explanation of its consequence. We denote any additions that go beyond the tradition separately (46 additions). All in all, 60 of the 67 approaches yield 198 approach-concept assessments (Tables 17 and 18).

Field	Code	Assigned when
Presence	Present	the approach fully implements the concept's defining function
	Partial	the approach implements only part of that function, or somehow a weaker form of it
	Absent	no part of the approach implements the function
Claimed	Yes	the source names the concept or its defining function and states that its proposal implements it
	Implied	the source does not name the concept, but entails it in the framing (e.g., through borrowed vocabulary or a claimed benefit that only holds if the concept does)
	No	the source neither states nor entails the concept (including concepts discussed only as background, recommendations to the field, or discussions of said concepts in other systems)

Table 16 | **Translation codes.** We read claims off the approach's own sources only, and never off what the tradition would require.

	Presence			
Claimed	Present	Partial	Absent	**Total**
Yes	10	33	1	44
Implied	0	49	3	52
No	0	28	74	102
Total	10	110	78	198

Table 17 | **Claimed × presence.** Shortfalls (claimed or implied, but not present): 86. Omissions (not claimed): 102.

Concept (source)	Presence	Claimed	Basis
Constitutional AI (Bai et al., 2022): "Training the constitution into the weights" (Constitutionalism)			
Priority among rules (cf. rule of recognition (Hart, 1961))	Absent	No	The 2022 principles form a flat list with no rule for which one wins when, e.g., helpfulness and harmlessness collide; Claude's 2026 constitution adds a priority order (Anthropic, 2026a)
Deciding borderline cases (cf. ~~Open~~open texture (Hart, 1961)~~)~~	Absent	No	Borderline cases have neither ~~a procedure nor a designated decider~~a designated decider nor a way to review how they were decided
Principles of legality (Fuller, 1964)	Partial	Implied	The principles are general and, once published, public, but whether behavior matches them is never checked
Service conception of authority (Raz, 1986)	Absent	No	The constitution is written in the lab, and whether deferring to it serves users' own reasons is never tested
Additions	-	-	A self-critique training loop in which one document both trains and evaluates the model
Preference aggregation: "Writing a voting rule into the training objective" (Deliberative aggregation)			
Deliberation, not tallying (Habermas, 1996)	Absent	No	One tallying rule is replaced by another, and reasons are never exchanged

Continued on the next page

Concept (source)	Presence	Claimed	Basis
General will vs. sum of preferences (Rousseau, 1997)	Absent	No	The inputs are private comparisons, and the methods differ only in how they weigh them
Veil of ignorance (Rawls, 1971)	Partial	Implied	Maximin builds the veil's output into the loss, but the objective is still imposed by the designer
Reasonable rejection (Scanlon, 1998)	Partial	No	The worst-off group sets the objective, but this group is a latent cluster rather than a person
Additions	-	-	Impossibility results serve as design constraints, and the aggregation rule itself becomes an optimization target
Collective deliberation (Huang et al., 2024): "Handing the decision to a convened public" (Deliberative aggregation)			
Deliberation, not tallying (Habermas, 1996)	Partial	Implied	Participants respond to each other's statements, but every procedure still decides by score
General will vs. sum of preferences (Rousseau, 1997)	Partial	Implied	Bridging-based ranking does not simply follow the raw majority, but it still only intersects private positions
Veil of ignorance (Rawls, 1971)	Absent	No	Sortition changes who chooses, but leaves what the chooser knows untouched
Reasonable rejection (Scanlon, 1998)	Partial	No	Per-side agreement floors give a veto to a bloc, but never to an individual
Additions	-	-	A pipeline runs from public input to the weights, and sortition is used to assemble a rater pool rather than a deliberating body

Table 18 | **Extraction examples** for the approaches behind the three case studies in Section 4. Each row corresponds to one approach-concept assessment, with the basis column paraphrasing the explanation recorded by the coder. As the baseline for preference aggregation, we use standard RLHF (Christiano et al., 2017; Ouyang et al., 2022). Since RLHF itself fails criterion I2, it is only cited as background (Section B.1).

B.9. Inter-annotator Agreement

Task	Unit	Double-coded	Agreement	Cohen's κ	Categories
Locus	Approach	20	100.0%	1.00	5
Disposition	Approach	20	85.0%	0.77	3
Kind	Record	30	86.7%	0.83	7
Presence	Approach × concept	30	86.7%	0.78	3
Claimed	Approach × concept	30	80.0%	0.69	3
Relevance	Record	single-coded	-	-	2
Inclusion	Record	200	86.5%	0.70	2
Kind, regime (screen)	Record	single-coded	-	-	7; 15
Keywording	Record	single-coded	-	-	-
Authority	Method record	single-coded	-	-	4
Authority fixed	Method record	single-coded	-	-	2

Table 19 | **Agreement between the two coders** before reconciliation (Cohen, 1960). We drew the double-coded items at random (20 approaches spanning all five loci and all three dispositions, 30 records stratified by kind, and 30 approach-concept pairs). Each coder was presented with the title, summary, and full text of each item assigned to them. They coded it on their own, without LLM involvement and access to the other coder's responses. Where the two coders disagreed (17 instances), the code was settled through discussion. Inclusion was checked in a separate coding process. A human screener who had not seen the original decisions was presented with 200 randomly sampled records (both included and excluded) and tasked with deciding their inclusion. This was compared to the LLM label (Table 7).

B.10. Corpus Statistics

			Records by kind								Translation	
Regime	Disposition	Appr.	Me	Be	An	Ap	Pr	Re	Pc	All	Conc.	Assess.
Codified												
Constitutionalism	Collapse	6 (6)	290	38	72	59	116	33	8	616	4	24
Contestatory constitutionalism	Contest	7 (2)	24	1	10	6	31	4	5	81	3	18
Consociational codification	Partition	7 (3)	94	23	26	15	46	19	15	238	4	21
Case-accumulated												
Precedential adjudication	Collapse	3 (3)	51	2	19	1	16	4	6	99	9	12
Divergent precedent	Contest	3 (3)	29	11	15	1	5	9	7	77	3	9
Legal pluralism	Partition	4 (2)	50	2	10	4	6	9	6	87	3	6
Customary												
Social convention	Collapse	4 (1)	50	17	64	15	3	4	0	153	5	9
Contested custom	Contest	2 (0)	13	3	5	3	5	0	5	34	3	6
Customary pluralism	Partition	5 (1)	80	122	240	37	20	36	3	538	4	15
Dispositional												
Virtue ethics	Collapse	3 (1)	32	73	115	12	7	2	0	241	6	12
Practical wisdom	Contest	3 (2)	9	7	9	1	6	0	0	32	2	6
Role pluralism	Partition	5 (3)	105	33	67	14	11	9	0	239	3	12
Procedural												

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			Records by kind								Translation	
Regime	**Disposition**	**Appr.**	Me	Be	An	Ap	Pr	Re	Pc	**All**	Conc.	Assess.
Deliberative aggregation	Collapse	6 (5)	177	11	71	52	69	8	64	452	12	20
Adversarial proceduralism	Contest	5 (2)	23	4	21	8	22	3	9	90	9	16
Polycentric governance	Partition	4 (2)	6	1	1	3	13	0	15	39	3	12
Cross-cutting (not placed)		-	0	0	37	0	3	0	0	40	-	-
Total		67 (36)	1,033	348	782	231	379	140	143	3,056	73	198

Table 20 | **Corpus by regime**, grouped by locus. We count records over the corpus, but approaches and the translation columns over the encyclopedia. *Appr.*: approaches (built). Kinds: *Me*thod, *Be*nchmark, *An*alysis, *Ap*plication, *Pr*oposal, *Re*source, *P*recedent. *Conc.*: the regime's tradition concepts. *Assess.*: approach-concept assessments.

		Authority in the experiments				
Regime	**Methods**	Dev.	Inst.	Pub.	Sys.	**Fixed**
Codified						
Constitutionalism	14	13	0	0	1	1
Contestatory constitutionalism	2	1	0	1	0	0
Consociational codification	5	1	3	0	1	1
Case-accumulated						
Precedential adjudication	7	2	1	3	1	1
Divergent precedent	5	2	1	2	0	1
Legal pluralism	3	2	0	1	0	0
Customary						
Social convention	3	0	0	0	3	3
Contested custom	0	0	0	0	0	0
Customary pluralism	1	0	0	0	1	1
Dispositional						
Virtue ethics	1	1	0	0	0	0
Practical wisdom	3	2	0	0	1	0
Role pluralism	7	4	0	3	0	0
Procedural						
Deliberative aggregation	20	6	0	14	0	6
Adversarial proceduralism	6	6	0	0	0	0
Polycentric governance	3	1	0	2	0	1
Total	80	41	5	26	8	15

Table 21 | **Authority over norms** for the methods in the encyclopedia, by regime: who holds it in each paper's experiments (*Dev.*: developer; *Inst.*: external institution; *Pub.*: affected public; *Sys.*: system itself) and how many methods fix it (Section 3.3). The 15 methods that fix authority place it with the affected public (8) or with the system itself (7).

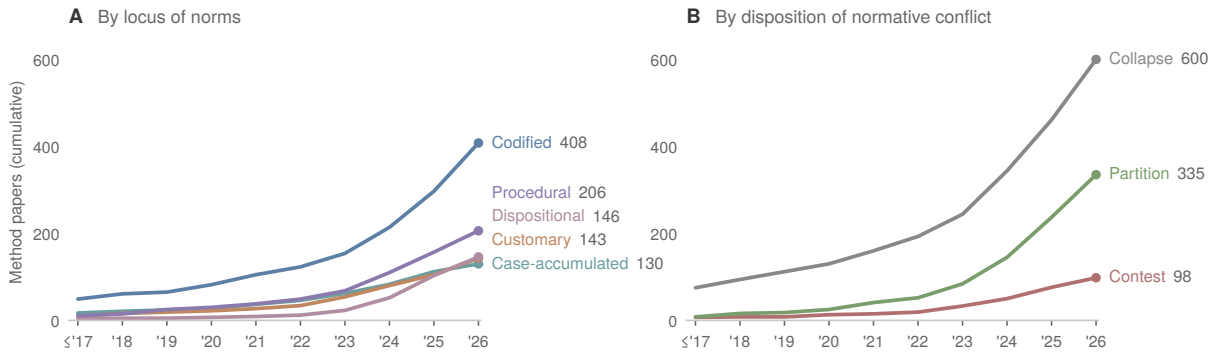


Figure 4 | **Growth of the Method Literature.** Cumulative number of method papers (in AI alignment and governance literature; the corpus of Section B.2) by year of first publication, grouped (A) by locus of norms and (B) by disposition of normative conflict. Work published before 2018 is pooled into the first year. Counts are within our corpus and are not normalized by the total number of publications per year; Table 22 gives the yearly counts and the share in the collapse column.

	Method records					
Year	Collapse	Contest	Partition	**All**	**Collapse share**	**All records**
≤2017	75	7	8	90	83%	228
2018	19	1	8	28	68%	65
2019	18	0	2	20	90%	65
2020	18	5	7	30	60%	78
2021	30	2	16	48	62%	113
2022	33	4	11	48	69%	132
2023	51	14	32	97	53%	264
2024	100	17	61	178	56%	458
2025	117	26	92	235	50%	733
2026	139	22	98	259	54%	920
Total	600	98	335	1,033	58%	3,056

Table 22 | **Method records by year of first publication** (the data behind Figure 4B, not cumulated), with the share in the collapse column and all records per year. 2026 runs to 26 August 2026, the cutoff of the systematic search. Years are those of first release: the first version's year for curated records, the arXiv identifier's year for other arXiv records, and otherwise the earliest year any index gives. Counts are within the corpus and not normalized by the field's total output per year.

C. Terminology

Term	Definition	Examples
Normative theory		
Tradition	A body of theory within a discipline.	Constitutionalism; deliberative democracy; legal pluralism
Concept	One constituent of a tradition.	Contestability; rule of recognition; amendment
Institution	A real-world arrangement that realizes a tradition's concepts.	Courts; amendment procedures; elections
Function	The role that a concept plays within its institution.	Appeal keeps decisions answerable to those affected by them
AI systems		
System	The AI system whose behavior is being governed.	A deployed LLM assistant
Case	A specific situation in which the system needs to act.	A request where being helpful would risk harm
Mechanism	A single technical component of a method.	A reward model or a critique-and-revision loop
Method	A concrete technical proposal consisting of one or more mechanisms.	Constitutional AI (Bai et al., 2022)
Approach	A group of methods that share the same design.	Debate before an under-informed judge
Regime	A combination of one locus of norms and one disposition of normative conflict.	Contestatory constitutionalism or adversarial proceduralism
Authority over norms	Who may write, interpret and change a system's norms. This is the third dimension of our property space. We code it by who supplies the norms in a paper's experiments.	The developer, an external institution, the affected public, or the system itself
Norms		
Norm	Any standard guiding a system's behavior, regardless of its form.	Rules, principles, precedents, or character traits
Rule	A norm that is explicitly stated ahead of time.	"Do not give instructions for building weapons."
Principle	A general norm that is part of a written specification.	"Choose the most helpful and honest response."
Normative translation		
Normative translation	How a method carries over the concepts of a tradition it draws on.	Constitutional AI as a translation of constitutionalism
Preserved / partially preserved / omitted	The degree to which a method carries over a concept: fully, partly, or not at all.	Constitutional AI preserves codification but omits amendment
Addition	Anything a method contributes beyond what its tradition provides.	Self-critique by the model
Shortfall	A concept the method claims to draw on but does not carry over.	Claiming contestability without offering any route to appeal
Translation gap	A concept of a tradition that is not carried over by the methods drawing on it.	Appeal in Constitutional AI
Coverage gap	A regime where there are few or no technical methods.	Contested custom